

Module 10: Multivalued Dependencies & 4NF

Multivalued Dependencies (MVD) & Fourth Normal Form (4NF)

Independent Multi-Valued Facts, Cartesian Tuple Multiplication & 4NF Decomposition

Formal Definition of Multivalued Dependency: $X \twoheadrightarrow Y$

In relation schema $R(X, Y, Z)$, $X \twoheadrightarrow Y$ holds iff for any two tuples t_1 and t_2 with $t_1[X] = t_2[X]$, there MUST also exist tuples t_3 and t_4 in R such that:

$t_3 = (t_1[X], t_1[Y], t_2[Z])$ and $t_4 = (t_1[X], t_2[Y], t_1[Z])$

The MVD Cartesian Explosion Hazard

Table: Course_Info(Course, Lecturer, TextBook)

Independent Facts Combine Multiplicatively:

- Course 'DBMS' has 2 Lecturers: {Dr. A, Dr. B}
- Course 'DBMS' has 2 Textbooks: {Korth, Navathe}

Requires $2 \times 2 = 4$ tuples in a single unnormalized table!

Why BCNF Cannot Catch This:

There are NO functional dependencies! Every attribute is part of the primary key: {Course, Lecturer, TextBook}.

Table is in BCNF, yet suffers from massive redundancy and update anomalies! We need 4NF!

Fourth Normal Form (4NF) Solution

Isolating Independent Multi-Valued Facts

Formal 4NF Condition:

Relation R is in 4NF iff it is in BCNF and for every non-trivial MVD $X \twoheadrightarrow Y$:

X MUST BE A SUPER KEY OF R .

4NF Decomposition Method:

Decompose $R(X, Y, Z)$ with $X \twoheadrightarrow Y$ into two schemas:

- $R_1(X, Y)$ with Key = $\{X, Y\}$
- $R_2(X, Z)$ with Key = $\{X, Z\}$

Decomposed relations: $2 + 2 = 4$ rows instead of cross product!

Lossless Join guaranteed by Fagin's Theorem!

> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Multivalued Dependencies (MVD: $\$X \twoheadrightarrow Y\$$), Limitations of Functional Dependencies, Independent Multivalued Facts, 4NF Definition, Fagin's Theorem, Solved AKTU Example.

1. Why Functional Dependencies Fall Short

Functional Dependencies ($\$X \rightarrow Y\$$)

only 1-to-1 ya many-to-1 associations ko model kar sakti hain.

Lekin real-world me aksar ek entity ke multiple independent attributes hote hain (1-to-many ya many-to-many). Jab do ya do se zyada independent multi-valued attributes ko ek hi relation me rakh diya jaata hai, toh **Cartesian Product Explosion** hota hai!

Is redundancy ko Functional Dependencies detect hi nahi kar paati kyunki koi attribute doosre ko uniquely determine nahi karta. Table ka poora attribute combination milkar Primary Key ban jaata hai, jisse relation BCNF satisfy kar leta hai par tab bhi corrupt aur redundant hota hai!

2. Multivalued Dependency (MVD: $\$X \twoheadrightarrow Y\$$)

2.1 Formal Definition

Relation $R(X, Y, Z)$ me jisme $Z = R - (X \cup Y)$, dependency:

$\$X \twoheadrightarrow Y\$$

Hold karti hai agar aur sirf agar kisi bhi relation instance me do tuples t_1 aur t_2 ke liye jahan $t_1[X] = t_2[X]$, relation me do aur tuples t_3 aur t_4 ka hona **anivarya (mandatory)** ho:

$$t_3 = (t_1[X], \text{quad } t_1[Y], \text{quad } t_2[Z])$$

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2.2 Hindi Intuition

Iska seedha matlab yeh hai ki attribute Y ki values jo X se judi hain, wo Z attribute ki values se **completely independent** hain!

Agar X ke paas m values of Y hain aur n values of Z hain, toh table me compulsory roop se m imes n tuples hone padenge!

3. Properties of Multivalued Dependencies

1. Complementation Rule:

- Agar $X \twoheadrightarrow Y$, toh $X \twoheadrightarrow (R - X - Y)$ bhi hamesha hold karega!
- (MVDs hamesha pairs me aati hain!).

2. Trivial MVD:

- Ek MVD $X \twoheadrightarrow Y$ trivial hoti hai agar:
- $Y \subseteq X$, ya fir
- $X \cup Y = R$.

3. FD implies MVD:

- Agar $X \rightarrow Y$, toh $X \twoheadrightarrow Y$ automatically true hota hai (MVD is a generalization of FD).

4. Fourth Normal Form (4NF) Formal Definition

A relation schema R is in **Fourth Normal Form (4NF)** with respect to a set of dependencies D (FDs and MVDs) if and only if it is in **BCNF**, and for every non-trivial Multivalued Dependency $X \twoheadrightarrow Y$ in D :

$$\mathbf{X \text{ ext } \{ \} R} \text{ is a SUPER KEY of } R$$

4.1 Fagin's Theorem (4NF Decomposition)

Agar relation $R(X, Y, Z)$ me MVD $X \twoheadrightarrow Y$ hold karti hai aur X super key nahi hai, toh R ko do relations me decompose kiya ja sakta hai:

$$R_1(X, Y) \text{ ext \{and\} } R_2(X, Z)$$

Yeh decomposition hamesha **Lossless Join** hota hai!

5. Architectural Diagram

6. Gateway Classes Solved AKTU Examination Problem

Problem: Relation STUDENT_INFO(Roll_No, Hobby, Language)

Assume ek student ke multiple hobbies aur multiple known languages ho sakte hain, aur hobby aur language ke beech koi direct relation nahi hai.

1. Identify Dependencies:

- $\text{Roll_No} \twoheadrightarrow \text{Hobby}$
- $\text{Roll_No} \twoheadrightarrow \text{Language}$

2. Current Keys & Normal Form:

- No attribute determines another. Candidate Key = $\{\text{Roll_No}, \text{Hobby}, \text{Language}\}$.
- All FDs trivial $\implies$ Relation is in **BCNF**!

3. Check 4NF:

- In $\text{Roll_No} \twoheadrightarrow \text{Hobby}$, the determinant Roll_No is **NOT a Super Key**!
- Therefore, relation **FAILS 4NF**!

4. 4NF Decomposition:

- $R_1(\text{Roll_No}, \text{Hobby})$
- $R_2(\text{Roll_No}, \text{Language})$

5. **Result:** Both R_1 and R_2 are in 4NF, redundancy is eliminated, and join is lossless!